



MUNICH · EDDM · 1.28" ROUND INSTRUMENT

Spotter's Field Manual

A pocket briefing for the little round radar on your desk. What every page shows, what each colour, triangle and number means, and how to drive it like you built it. Keep it by the device and glance across whenever the sky gets interesting.

6 live pages

auto-rotate + button

live ADS-B traffic

decoded METAR

rare-aircraft alerts

00

ORIENTATION

What this thing actually is

It is a standalone desk instrument that watches the real sky around you and around Munich Airport. There is no phone app and no companion server — the ESP32 microcontroller talks straight to free public flight and weather feeds over your Wi-Fi, and paints the result on a 240 × 240 round screen. Three feeds do all the work:

FEED	GIVES YOU	HOW OFTEN
adsb.lol	Live aircraft positions (ADS-B) around home & MUC	every 8 s
adsbdb	Route (origin → destination) & airline for tracked callsigns	on demand
aviationweather.gov	Decoded Munich (EDDM) METAR weather	every 5 min

Between the 8-second position updates the device gently **dead-reckons** — it nudges each aircraft forward along its own heading and speed so motion looks continuous instead of teleporting. Nothing you see is faked; it is just smoothing the gap between refreshes.

01

CONTROLS

Cycling the pages & reading the clock

Left to itself, the device runs an **auto-carousel**: it dwells on each page for a set time, then slides to the next. One button changes that.

THE BUTTON

short press → jump to the next page **and freeze there**. The carousel stops so you can stare as long as you like.

hold ≈1.5 s → **resume** the automatic carousel from wherever you are.

It is the onboard button on the microcontroller. Presses are caught even while the device is mid-download, so it never feels stuck.

AUTO DWELL TIMES

Each page stays up for a tuned interval, longest where there is most to read:

- Home Radar — **26 s**
- MUC Map — **17 s**
- Traffic Brief — **16 s**
 - Nearest — **22 s** · Coolest — **22 s**
- MUC Weather — **13 s**

Interrupts: the device can override the carousel on its own. An **emergency squawk** (7700/7600/7500) or a very rare aircraft (coolness score ≥ 95) yanks the display straight to the traffic/coolest page so you never miss it.

02

THE SIX PAGES

Page by page

The carousel order is fixed. Here is exactly what each screen is telling you.



1 HOME RADAR

north-up scope, centred on you

Your world from above, north at the top, your position at the crosshair. Every aircraft is drawn where it actually is relative to you.

- ▶ quietly tour
20 / 40 / 60 km — the label at the bottom (e.g. 40 km) tells you the current outer ring. Busier skies favour a tighter zoom so labels stay readable.

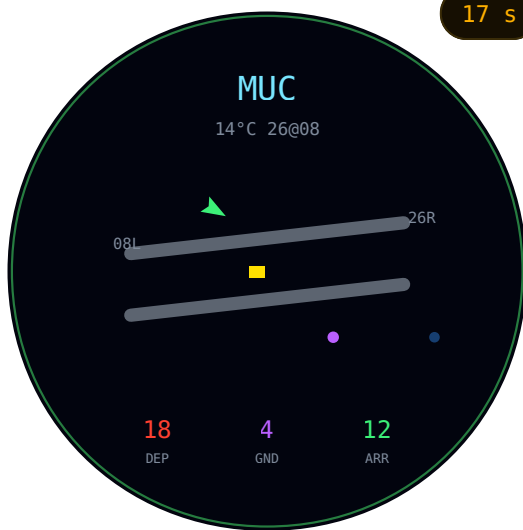
-
- ▶ white
under the red north arrow, red on the west side, green on the east.

-
- ▶ ■ ■ ■
=
arriving-at-MUC /
departing-MUC / other
traffic. Each points along
its heading.

-
- ▶ ■ = an
aircraft beyond the drawn
ring — direction only,
deliberately dim so a ring

of distant traffic doesn't shout.

- ▶ The small grey dot dead-centre is (the receiver).



2 MUC MAP

airport operations view

Munich Airport drawn as a schematic — the two real parallel runways, 08L/26R and 08R/26L, at true geometry.

- ▶ the MUC title, then live temperature + wind pulled from the METAR, then the DEP / GND / ARR counters along the bottom.

- ▶ airborne traffic is mapped on a fixed 20 km view (contacts from 20–40 km become blue rim markers); aircraft actually *on the field* are drawn at runway scale so ground movements spread out realistically.

- ▶  = an aircraft physically on a runway. A bigger slab means a heavier jet.

- ▶  = on the ground / taxiing.

- ▶ Labels ride on the next arrival and next departure: callsign + fr XXX / to XXX, then the arrival's ETA

or the departure's distance
out.

16 s



3 TRFC

the traffic brief

A one-glance summary of the four aircraft worth knowing about right now, plus a mood-gauge of how busy the wider sky is.

- ▶ ■ — the closest airborne aircraft to you.
- ▶ ■ — the best-scored special right now (rare type, military, government...).
- ▶ ■ — the nearest helicopter, with its operator role if known (police, EMS...).
- ▶ ■ — anything squawking an emergency code. Usually "— none —", and that's good.
- ▶ The footer counts every valid aircraft within of you — a quick read on whether the sky is dead or heaving.



4 NEAREST

tracking card · closest plane

The full data card for whatever is closest overhead, refreshed every second so the numbers breathe.

- ▶ big at the top, then in the airline's brand colour.

- ▶ as ICAO codes and full city names, plus a note if it's a Munich arrival or departure.

- ▶ Live (altitude, ft), (ground speed, kt), (distance to you, km).

- ▶ A in metres-per-minute and the — "passes 2.1 km from you in 1m36s", the closest it will get.

- ▶ The points along the bearing *from you* to the plane. Face it, look up — that's where it is.

22 s



5 COOLEST

tracking card · best score

Same card format as Nearest, but locked onto the most *interesting* aircraft in range rather than the closest — with a one-line reason it made the cut.

- ▶ Picks the highest A400M ATLAS in the sky: superjumbos, military transports, government flights, emergencies and more (full table below).

-
- ▶ The A400M ATLAS, QUEEN B747, FIGHTER JET spells out the reason — e.g. A400M ATLAS, QUEEN B747, FIGHTER JET.

-
- ▶ Anything scoring 93 interrupts the carousel and jumps straight here, once per callsign, so a genuine rarity can't slip past.

-
- ▶ Same red rim bearing blip — turn and look.

13 s

MUC WX

WIND	260° 8kt
VIS	10 km
CLOUD	BKN 1800ft
TEMP	14 / 09°C
QNH	1017 hPa

VFR · rwy 26

EDDM 260806Z 26008KT 9999...

6 MUC WX

decoded Munich weather

The airport's official METAR observation, decoded into plain rows instead of the raw code — refreshed every 5 minutes.

- ▶ direction + speed, visibility, lowest reported layer & base, temperature/dewpoint, the altimeter pressure setting.
- ▶ A row flags significant weather (rain, snow, fog...) — "DRY" when the sky is clean.
- ▶ A best-guess from the wind (e.g. landing on 26 into a westerly).
- ▶ The summarises flying conditions: (green, clear) through MVFR and IFR down to (red, very low). See the glossary.
- ▶ The tiny grey line at the bottom is the for purists.

Symbol & colour grammar

One consistent visual language runs across every page. Learn these once and you can read any screen at a glance.



Green triangle

Arriving at MUC — points along its heading



Red triangle

Departing MUC



Amber triangle

Other traffic (not tied to MUC)



Yellow slab

On a runway — bigger = heavier aircraft



Purple dot

On the ground / taxiing



Faint blue rim dot

Beyond the drawn scope — direction only



Star

Special / rare aircraft (scored ≥ 70)



Rotor cross

Helicopter



Red "!" pill

Emergency squawk 7700 / 7600 / 7500



Red rim blip

On tracking pages: bearing from YOU to the plane

COUNTER COLOURS

5

Red number

Departures

7

Green number

Arrivals

4

Purple number

On the ground

12

White number

Total contacts

The coolness score

The Coolest page and the star symbol are driven by a scoring system. Aircraft type and callsign each earn a score; the highest wins. ≥ 70 earns a star; ≥ 95 interrupts the carousel. A few "live situation" bonuses can also promote an ordinary plane doing something interesting.

BY AIRCRAFT TYPE

SCORE	CALLED OUT AS	SCORE	CALLED OUT AS
100	Superjumbo A380	92	Queen B747 / 747-8
99	Antonov An-124 · B-52 bomber	86	Rare trijet MD-11
97	C-5 Galaxy	72	Rare quad A340-500/600
96	C-17 Globemaster · Eurofighter	55	Big twin B777
95	F-16 · F-35 fighter jet	52	A350-1000
93	A400M Atlas	45	Dreamliner B787
90	C-130 Hercules · Ilyushin Il-76	38	Airbus A330

BY CALLSIGN PREFIX

SCORE	PREFIX	MEANING
100	AF1	Air Force One?!
98	SAM	US Government special
95	GAF · NATO	German Air Force · NATO flight
92	RRR	Royal Air Force
88	CTM · MMF	French Air Force · NATO tanker
86	RCH	US Air Force transport
85	DUKE	US Army

LIVE SITUATION BONUSES

SCORE	SITUATION
82	Low overhead pass — within 4 km, below 3 500 ft
78	Fast low traffic — over 430 kt below 20 000 ft
70	MUC final inbound — descending toward the airport
66	MUC climbout — climbing away just after departure

Emergency squawks trump everything. A transponder set to 7700 (general emergency), 7600 (radio failure) or 7500 (unlawful interference) scores 100 outright and is flagged with the red "!" pill.

Terms & jargon

Automatic Dependent Surveillance—Broadcast. Aircraft broadcast their own GPS position, altitude and ID; this device reads a free network of receivers, not a radar dish.

The flight's radio ID, e.g. DLH4AB. The first three letters are the airline (DLH = Lufthansa). Not the same as the tail/registration.

A 4-digit transponder code. Most are routine; 7500/7600/7700 are the emergency codes.

A coded airport weather observation, reissued about every 30 minutes. This device decodes Munich's into plain rows.

The altimeter pressure setting (hPa) that makes an altimeter read height above sea level. Rising QNH ≈ improving weather.

Height of the lowest broken/overcast cloud layer — the "lid" pilots care about.

How far you can see horizontally; "9999" or "10 km" means 10 km or more.

Flight categories by ceiling & visibility, best to worst: Visual, Marginal Visual, Instrument, Low Instrument.

Colour-coded green → red.

A compact aircraft-model code, e.g. A20N = Airbus A320neo, B77W = Boeing 777-300ER.

Two naming systems: MUC (IATA, 3-letter) and EDDM (ICAO, 4-letter) are the same airport.

Heading = where the nose points; track = the path actually made good over the ground. The triangles point along track.

Speed over the ground in knots (1 kt ≈ 1.85 km/h).

Height in feet; airliners cruise around 30 000–40 000 ft.

Climb/descent speed. Shown here as a climb trend in metres per minute (mpm).

Closest Point of Approach — how near a plane will pass you, and when, if both of you hold course.

Numbers are the runway's magnetic heading $\div 10$ (08 $\approx$ 080°); the same strip is 26 from the other end. L/R distinguishes parallel runways.

Estimating a position between real updates by projecting last known speed & heading forward — how the traffic keeps moving between 8-second fetches.

The compass direction from you to a target. The red rim blip on tracking pages is a live bearing.

06

TEAR-OFF

Desk cheat-card

PAGE ORDER

1 · Home Radar 2 · MUC Map 3 · Traffic
4 · Nearest 5 · Coolest 6 · MUC WX

BUTTON

tap next page + freeze
hold resume auto-rotate

TRIANGLES

■ arriving ■ departing ■ other

DOTS & MARKS

■ on runway ■ taxiing
■ off-scope ★ rare ! emergency

STARS APPEAR AT

score ≥ 70 · alert jump at ≥ 95